



CLASSROOM CONTACT PROGRAMME

TG: @Chalnaayaaar (Academic Session : 2023-2024)

NEET(UG)

MINOR

23-04-2023

PRE-MEDICAL : ENTHUSIAST ADVANCE COURSE PHASE : MEF-I (B)

IMPORTANT NOTE : Students having 8 digits **Form No.** must fill two zero before their Form No. in OMR. For example, if your **Form No.** is 12345678, then you have to fill **0012345678**.

Test Booklet Code

This Booklet contains 24 pages.

E2

Do not open this Test Booklet until you are asked to do so.

Important Instructions :

- The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on ORIGINAL Copy carefully with **blue/black** ball point pen only.
- The test is of **3 hours 20 minutes** duration and the Test Booklet contains **200** multiple-choice questions (four options with a single correct answer) from **Physics, Chemistry and Biology (Biology-I and Biology-II)**. 50 questions in each subject are divided into **two Sections (A and B)** as per details given below :
 - Section A** shall consist of **35 (Thirty-five)** Questions in each subject (Question Nos - 1 to 35, 51 to 85, 101 to 135 and 151 to 185). All questions are compulsory.
 - Section B** shall consist of **15 (Fifteen)** questions in each subject (Question Nos - 36 to 50, 86 to 100, 136 to 150 and 186 to 200). In Section B, a candidate needs to **attempt any 10 (Ten)** questions out of **15 (Fifteen)** in each subject. **Candidates are advised to read all 15 questions in each subject of Section B** before they start attempting the question paper. In the event of a candidate attempting more than ten questions, **the first ten questions answered by the candidate shall be evaluated.**
- Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, **one** mark will be deducted from the total scores. **The maximum marks are 720.**
- Use **Blue/Black Ball Point Pen only** for writing particulars on this page/markings responses on Answer Sheet.
- Rough work is to be done in the space provided for this purpose in the Test Booklet only.
- On completion of the test, the candidate **must hand over the Answer Sheet (ORIGINAL and OFFICE Copy) to the Invigilator** before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
- The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Form No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
- Use of white fluid for correction is **NOT** permissible on the Answer Sheet.
- Each candidate must show on-demand his/her Allen ID Card to the Invigilator.
- No candidate, without special permission of the Invigilator, would leave his/her seat.
- The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign (with time) the Attendance Sheet **twice**. **Cases, where a candidate has not signed the Attendance Sheet second time, will be deemed not to have handed over the Answer Sheet and dealt with as an Unfair Means case.**
- Use of Electronic/Manual Calculator is prohibited.
- The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Room/Hall. All cases of unfair means will be dealt with as per the Rules and Regulations of this examination.
- No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.**
- The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.
- Compensatory time of one hour five minutes will be provided for the examination of three hours and 20 minutes duration, whether such candidate (having a physical limitation to write) uses the facility of scribe or not.

Name of the Candidate (in Capitals) : _____

Form Number : in figures _____

: in words _____

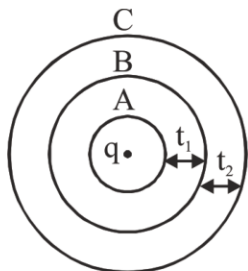
Centre of Examination (in Capitals) : _____

Candidate's Signature : _____ Invigilator's Signature : _____

Your Target is to secure Good Rank in Pre-Medical 2024

SECTION-A (PHYSICS) TG: @Chalnaayaaar

1. Figure shows three spherical and equipotential surfaces A, B and C round a point charge q . The potential difference $V_A - V_B = V_B - V_C$. If t_1 and t_2 be the distances between them, then :-

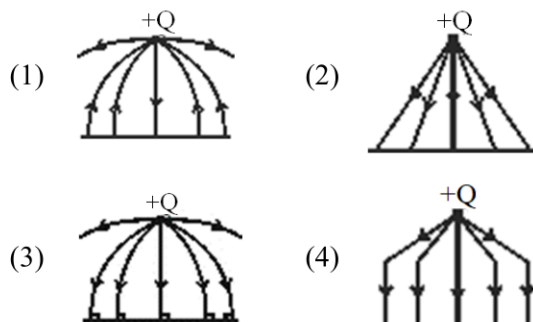
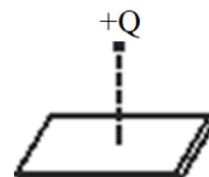


- (1) $t_1 = t_2$ (2) $t_1 > t_2$ (3) $t_1 < t_2$ (4) $t_1 \leq t_2$
2. If we have an irregularly shaped conductor as shown and a charge is given to it, choose the correct statement.

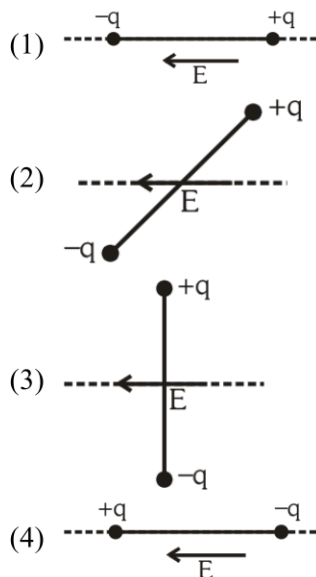


- (1) the electric field and electric potential at P is maximum among all points of conductor.
- (2) the electric field at P is maximum and electric potential is same at all points of conductor.
- (3) the electric field and electric potential at P is same at all points of conductor.
- (4) the electric field and electric potential at P is minimum among all points of conductor.
3. A big hollow metal sphere A is charged to V volts and another smaller hollow sphere B is charged to $V/2$ volts. If B is put inside A and joined with a metallic wire, then the direction of charge flow :-
- (1) is from A to B.
- (2) is from B to A.
- (3) no charge flows.
- (4) depends on the radii of spheres.

4. A charge Q is fixed at a distance d in front of an infinite metal plate. The lines of force are represented by :-



5. In which of the following states is the potential energy of an electric dipole maximum -



6. A charged particle of mass m and charge q is released from rest in a uniform electric field E . Neglecting the effect of gravity, the kinetic energy of the charged particle after ' t ' second is :-

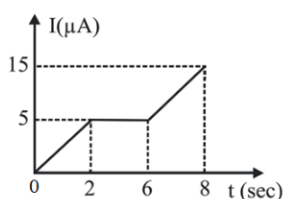
(1) $\frac{Eq^2m}{2t^2}$ (2) $\frac{2E^2t^2}{mq}$

(3) $\frac{E^2q^2t^2}{2m}$ (4) $\frac{Eqm}{t}$

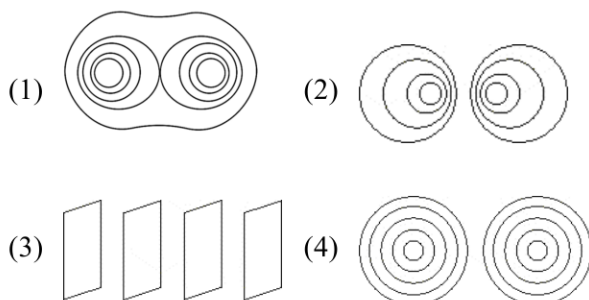
7. Six charges $+Q$ each are placed at each corner of a regular hexagon. Find the total potential energy :-

(1) $\frac{6kQ^2}{r} + \frac{3kQ^2}{2r} + \frac{6kQ^2}{\sqrt{3}r}$
 (2) $\frac{6kQ^2}{r} + \frac{6kQ^2}{2r}$
 (3) $12 \frac{kQ^2}{r}$
 (4) $\frac{6kQ^2}{r} + \frac{3kQ^2}{2r} + \frac{3kQ^2}{\sqrt{3}r}$

8. I-t curve is shown in figure. Calculate total charge flown in circuit from $t = 0$ to $t = 8$ sec.



- (1) $35 \mu\text{C}$ (2) $40 \mu\text{C}$
 (3) $45 \mu\text{C}$ (4) None of these
9. **Statement-1** : Potential energy of a charge q at any point (in the presence of field due to any charge configuration) is the work done by the external force (equal and opposite to the electric force) in bringing the charge q from infinity to that point.
Statement-2 : Electric field is in the direction in which the potential decreases steepest.
- (1) Statement-1 is correct statement-2 is incorrect
 (2) Statement-1 is incorrect statement-2 is correct
 (3) Statement-1 and 2 both are correct
 (4) Statement-1 and 2 both are incorrect
10. Which of the following most correctly represents equipotential surfaces of 2 identical positive charges?



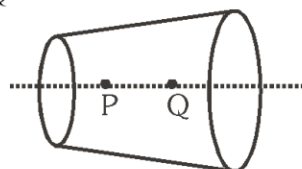
11. Choose the incorrect statement for conductors in static situation.

- (1) The interior of a conductor can have no excess charge
 (2) Electrostatic field is zero inside a conductor
 (3) Electrostatic potential is constant throughout the volume of the conductor and has the same value (as inside) on the surface.
 (4) Electric field at the surface of a charged conductor is $\vec{E} = \frac{\sigma}{2\epsilon_0} \hat{n}$ where σ is surface charge density and $\hat{n}$ is unit vector normal to the surface outward

12. A regular hexagon of side 10 cm has a charge $5\mu\text{C}$ at each of its vertices. Calculate the potential at the center of the hexagon.

- (1) $2.7 \times 10^6 \text{ V}$ (2) $2.7 \times 10^{12} \text{ V}$
 (3) $8.7 \times 10^6 \text{ V}$ (4) $8.7 \times 10^{12} \text{ V}$

13. A wire has a non-uniform cross-section as shown in figure. A steady current flows through it. The drift speed of electrons at points P and Q is v_P and v_Q , then :-



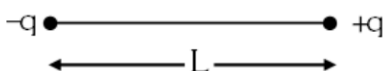
- (1) $v_P = v_Q$ (2) $v_P < v_Q$
 (3) $v_P > v_Q$ (4) data is insufficient
14. The reciprocal of resistance is :

- (1) reactance (2) mobility
 (3) conductivity (4) conductance

15. When a particle with charge $+q$ is thrown with an initial velocity v towards another stationary charge $+Q$, it is repelled back after reaching the nearest distance r from $+Q$. The closest distance that it can reach if it is thrown with initial velocity $2v$, is:

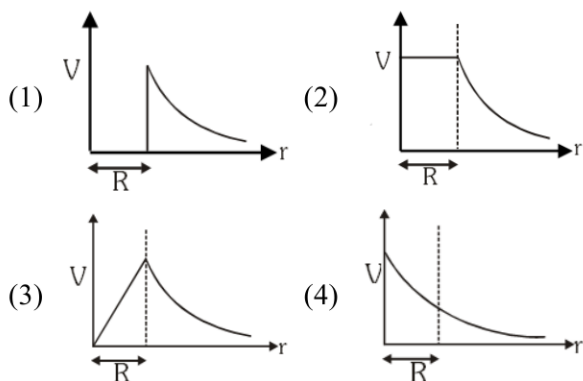
- (1) $\frac{r}{2}$ (2) $\frac{r}{16}$ (3) $\frac{r}{8}$ (4) $\frac{r}{4}$

16. Two point charges $-q$ and $+q$ are placed at a distance of L , as shown in the figure.



The magnitude of electric field intensity at a distance R ($R \gg L$) varies as :

- (1) $\frac{1}{R^3}$ (2) $\frac{1}{R^4}$ (3) $\frac{1}{R^6}$ (4) $\frac{1}{R^2}$
17. The variation of electrostatic potential with radial distance r from the centre of a positively charged metallic thin shell of radius R is given by the graph



18. An unnumbered wall clock shows time 8 : 20, what time will its image in a plane mirror shows:-

(1) 3 : 40 (2) 4 : 40 (3) 5 : 20 (4) 4 : 20

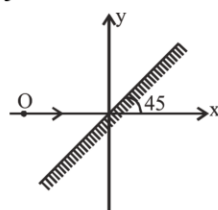
19. In front of a concave mirror an object is placed at a distance x from the focus, and the real image is formed at a distance y from the focus. The focal length of the mirror is :-

(1) xy (2) $\sqrt{xy}$ (3) $x+y$ (4) $\sqrt{\frac{x}{y}}$

20. Two plane mirror are placed at an angle of 60° . Calculate no. of image formed if an object is placed in between them

(1) 6 (2) 5 (3) 4 (4) 3

21. Image of the object can be :-



(1) $\downarrow$ (2) $\uparrow$ (3) $\rightarrow$ (4) $\nwarrow$

22. A concave mirror of focal length f (in air) is immersed in water ($\mu = 4/3$). The focal length of mirror in water will be :-

(1) f (2) $(4/3)f$
(3) $(3/4)f$ (4) $(7/3)f$

23. A 2.0 cm tall object is placed 15 cm in front of a concave mirror of focal length 10 cm. What is the size and nature of the image :-

(1) 4 cm, real (2) 4 cm, virtual
(3) 1.0 cm, real (4) None of these

24. A concave mirror forms the image of an object on a screen. If the lower half of the mirror is covered with an opaque card, the effect would be :-

(1) To make the image less bright
(2) to make the lower half of the image
(3) to make the upper half of the image disappear
(4) to make the image blurred

25. A short linear object of length 1 mm lies on the axis of a concave mirror of focal length 10 cm at a distance 15 cm from the mirror. its image has an axial length equal to :-

(1) 4 mm (2) 2mm
(3) 0.25 mm (4) 0.5 mm

26. A ray of light incident on a plane mirror at an angle of incidence of 30° The deviation produced by the mirror is :

(1) 30° (2) 60° (3) 90° (4) 120°

27. A plane mirror is moving with velocity $4\hat{i} + 5\hat{j} + 8\hat{k}$. A point object in front of the mirror moves with a velocity $3\hat{i} + 4\hat{j} + 5\hat{k}$. here $\hat{k}$ is along the normal to the plane mirror and facing towards the object. The velocity of the image is :

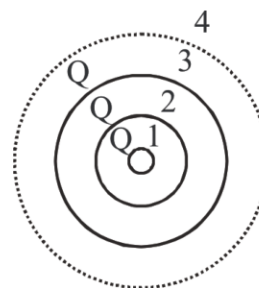
(1) $-3\hat{i} - 4\hat{j} + 5\hat{k}$ (2) $3\hat{i} + 4\hat{j} + 11\hat{k}$
(3) $-3\hat{i} - 4\hat{j} + 11\hat{k}$ (4) $7\hat{i} + 9\hat{j} + 11\hat{k}$

28. An object is placed asymmetrically between two plane mirrors, inclined at an angle of 72° . Find the total number of images formed.
- (1) 5 (2) 4 (3) 2 (4) Infinite
29. Two plane mirrors are inclined to each other at an angle of 60° . If a ray of light incident on the first mirror is parallel to the second mirror, it is reflected from the second mirror :
- (1) Perpendicular to the first mirror.
 (2) Parallel to the first mirror.
 (3) Parallel to the second mirror
 (4) Perpendicular to the second mirror.
30. In case of concave mirror, the minimum distance between a real object and its real image is :
- (1) f (2) $2f$ (3) $4f$ (4) zero
31. Which of the following can form erect, virtual, diminished image ?
- (1) Plane mirror (2) Concave mirror
 (3) Convex mirror (4) Both (2) & (3)
32. The focal length of a concave mirror is 20 cm. Determine where an object must be placed to form an image magnified two times when the image is real -
- (1) 30 cm from mirror
 (2) 10 cm from mirror
 (3) 20 cm from mirror
 (4) 15 cm from mirror
33. An object is 24 cm above the surface of a lake. At what distance from the surface, a fish will see this object ? ($\mu_{\text{water}} = 4/3$)
- (1) 32 cm above surface
 (2) 18 cm above surface
 (3) 18 cm below surface
 (4) 32 cm below surface

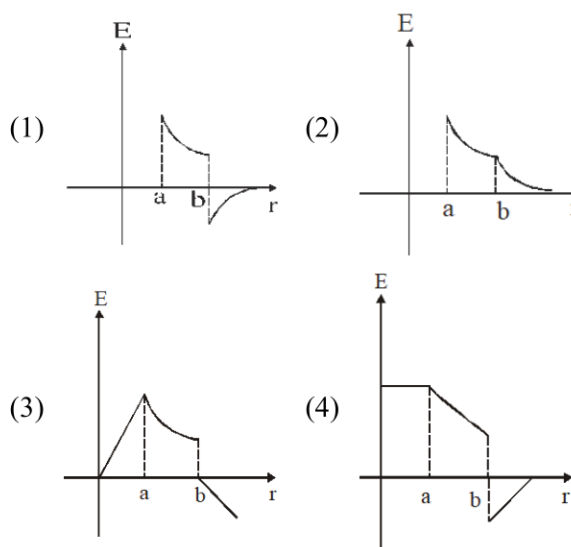
34. Time taken by light to cross a 4mm window glass of refractive index 1.5 will be:
- (1) 2×10^{-8} sec (2) 2×10^8 sec
 (3) 2×10^{-11} sec (4) 2×10^{11} sec
35. Wavelength of light in vacuum is $6000\text{\AA}$ & in a medium its $4000\text{\AA}$. Find refractive index of the medium.
- (1) 2.4 (2) 1.5 (3) 1.2 (4) 0.67

SECTION-B (PHYSICS)

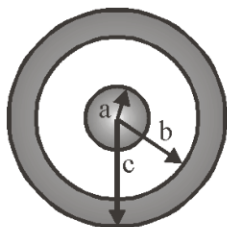
36. An infinite number of concentric rings carry a charge Q each alternately positive and negative. Their radii are 1, 2, 4, 8.....meters in geometric progression as shown in the figure. The potential at the centre of the rings will be :-



- (1) zero (2) $\frac{Q}{12\pi\epsilon_0}$ (3) $\frac{Q}{8\pi\epsilon_0}$ (4) $\frac{Q}{6\pi\epsilon_0}$
37. Two concentric conducting thin spherical shells of radii a and b ($b > a$) are given charges Q and $-2Q$ respectively. The electric field along a line passing through centre as a function of distance (r) from centre is represented by :-



38. A solid conducting sphere of radius a has a net positive charge $2Q$. A conducting spherical shell of inner radius b and outer radius c is concentric with the solid sphere and has a net charge $-Q$. The surface charge density on the inner and outer surfaces of the spherical shell will be :-



- (1) $-\frac{2Q}{4\pi b^2}, \frac{Q}{4\pi c^2}$
 (2) $-\frac{Q}{4\pi b^2}, \frac{Q}{4\pi c^2}$
 (3) $0, \frac{Q}{4\pi c^2}$
 (4) None of the above
39. A dipole of moment $P = 2 \text{ C m}$ is kept along $+y$ in field directed along $+y$ & increasing at rate of 10 V/m^2 . Field at location of dipole is $\vec{E} = 5 \text{ N/C} \hat{j}$.

	Column-I		Column-II
A	U	I	-10
B	τ	II	20
C	F	III	0

[All values in SI system]

- (1) A-II, B-II, C-I
 (2) A-I, B-III, C-II
 (3) A-I, B-II, C-III
 (4) A-III, B-II, C-I
40. In hydrogen atom the electron makes 6.6×10^{15} revolution per second around the nucleus in an orbit of radius $0.5 \times 10^{-10} \text{ m}$. It is equivalent to a current nearly.
- (1) 1 A
 (2) 1 mA
 (3) 1 μA
 (4) $1.6 \times 10^{-19} \text{ A}$

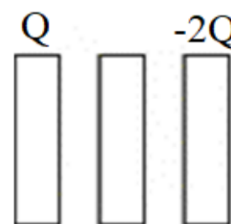
41. Inside a ball charged uniformly with volume density ρ , there is a spherical cavity. The centre of the cavity is displaced with respect to the centre of the ball by a distance $\vec{a}$. Find the field strength $\vec{E}$ inside the cavity.

- (1) $\frac{1}{3} \frac{\vec{a} \rho}{\epsilon_0}$ (2) $\frac{1}{2} \frac{\vec{a} \rho}{\epsilon_0}$
 (3) $\frac{1}{4} \frac{\vec{a} \rho}{\epsilon_0}$ (4) Zero.

42. The potential of a certain electrostatic field has the form $V = a(x^2 + y^2) + bz^2$ where a and b are constants. Find the magnitude and direction of the electric field strength vector.

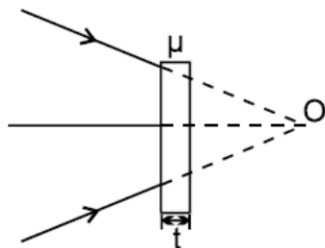
- (1) $\vec{E} = -2(ax\hat{i} + by\hat{j} + az\hat{k})$
 (2) $\vec{E} = -(ax\hat{i} + ay\hat{j} + bz\hat{k})$
 (3) $E = \sqrt{a^2(x^2 + y^2) + b^2z^2}$
 (4) $\vec{E} = -2(ax\hat{i} + ay\hat{j} + bz\hat{k})$

43. Three identical metal plates with large surface area are kept parallel to each other. The left most plate is given a charge Q , the rightmost a charge $-2Q$ and the middle one remain neutral. Find the charge on outer surface of rightmost plate.



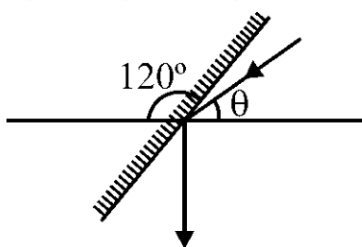
- (1) $\frac{-3Q}{4}$ (2) $\frac{+Q}{2}$
 (3) $\frac{-Q}{2}$ (4) $\frac{3Q}{4}$
44. A concave mirror is used to focus the image of a flower on a nearby wall 120 cm from the flower. If a lateral magnification of 16 is desired, the distance of the flower from the mirror should be
- (1) 8 cm (2) 12 cm
 (3) 80 cm (4) 120 cm

45. A beam of light is converging towards a point. A plane parallel plate of glass of thickness t , refractive index μ is introduced in the path of the beam. The convergent point is shifted by (assume near normal incidence):



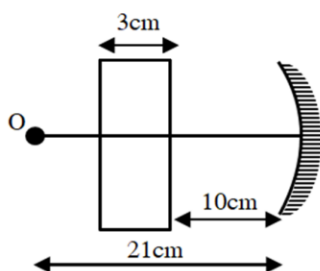
- (1) $t \left(1 - \frac{1}{\mu}\right)$ away
 (2) $t \left(1 + \frac{1}{\mu}\right)$ away
 (3) $t \left(1 - \frac{1}{\mu}\right)$ nearer
 (4) $t \left(1 + \frac{1}{\mu}\right)$ nearer

46. Find the angle θ in given diagram :-



- (1) 70° (2) 60° (3) 80° (4) 30°

47. An object is placed 21 cm in front of a concave mirror of radius of curvature 20 cm



A glass slab of thickness 3 cm and refractive index 1.5 is placed close to the mirror in space between the object and the mirror. Find the position of final image formed if distance of nearer surface of the slab from the mirror is 10 cm.

- (1) 20 cm (2) 21 cm (3) 19 cm (4) 40 cm

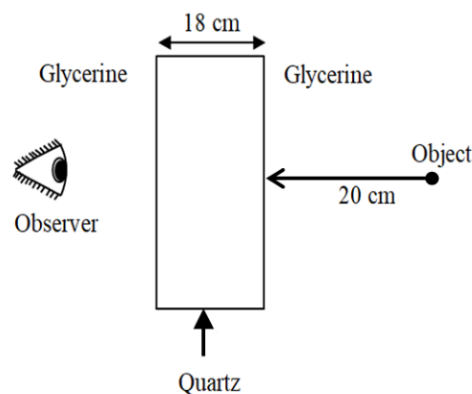
48. How much water should be filled in a container 21 cm in height, so that it appears half filled when viewed from the top of the container (given that ${}_a\mu_w = 4/3$) :-

- (1) 8.0 cm
 (2) 10.5 cm
 (3) 12.0 cm
 (4) None of the above

49. A bubble in a glass slab ($\mu=1.5$) when viewed from one side appears at 5 cm and at 2 cm from the other side. Find thickness of slab :

- (1) 3.75 cm
 (2) 23 cm
 (3) 10.5 cm
 (4) 1.5 cm

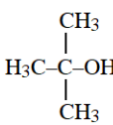
50. Given that velocity of light in quartz is 1.5×10^8 m/s & velocity of light in glycerine is $\frac{9}{4} \times 10^8$ m/s. Now, a slab made of quartz is placed in glycerine as shown. The shift of the object produced by slab is :



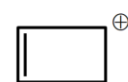
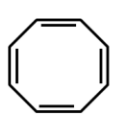
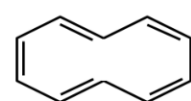
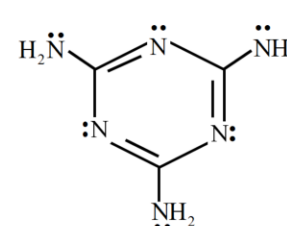
- (1) 6 cm
 (2) 3.55 cm
 (3) 9 cm
 (4) 2 cm

SECTION-A (CHEMISTRY)

51. Which of following is easiest to deprotonate ?

- (1)  (2) $\text{H}_3\text{C}-\text{NH}_2$
 (3) $\text{HC}\equiv\text{CH}$ (4) $\text{H}_3\text{C}-\text{CH}_2-\text{OH}$

52. Which of following is aromatic ?

- (1) 
 (2) 
 (3) 
 (4) 

53. Which of the following is/are correct statement(s) regarding CH_3^- carbanion ?

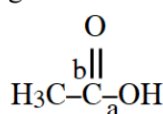
- (a) three b.p. and one l.p. in valence shell
 (b) four b.p. in valence shell
 (c) only three b.p. in valence shell [No other electrons]
 (d) sp^3 hybrid C-atom
 (e) sp^2 hybrid C-atom

- (1) a, b, d (2) a, d (3) c, d (4) b, d

54. Which is not nucleophile among the following :-

- (1) H_2O (2) CH_3-OH
 (3) AlCl_3 (4) $\text{CH}_2=\text{CH}_2$

55. Order of Bond length ?

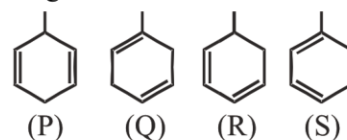


- (1) $a < b$ (2) $a > b$
 (3) $a = b$ (4) can't predict

56. Hyperconjugation is best described as :-

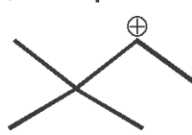
- (1) delocalisation of π electrons into a nearby empty orbital
 (2) delocalisation of σ electrons into a nearby empty orbital
 (3) Electron donating effect of alkyl group having a C-H σ bond, when present in conjugation with anion.
 (4) the migration of a carbon or hydrogen from one carbocation to another

57. Arrange the following in decreasing order of heat of hydrogenation :

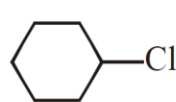
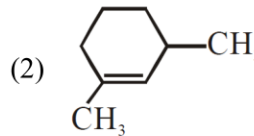
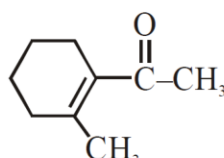


- (1) $P > Q > R > S$ (2) $S > R > Q > P$
 (3) $Q > P > S > R$ (4) $R > Q > P > S$

58. In which of the following compound hyperconjugation is observed ?

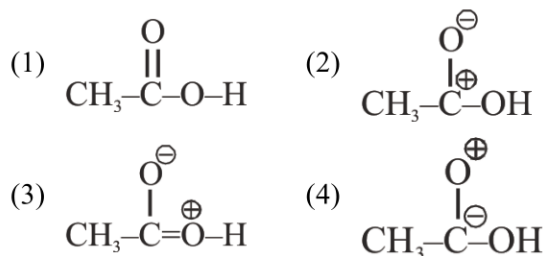
- (1) 
 (2) 
 (3) 
 (4) None of these

59. In which of the following molecules all the effects namely inductive, mesomeric and hyperconjugation operate ?

- (1)  (2) 
 (3)  (4) 



Acetic acid
A, B, C, D are resonating structure of acetic acid, structure of B is :-



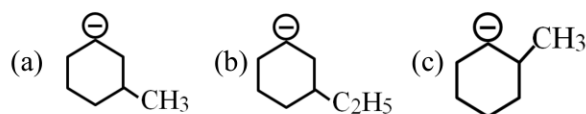
61. Select the incorrect statement :-

- (1) In CH_3^+ , the vacant orbital is perpendicular to the molecular plane.
- (2) Homolytic cleavage results in the formation of neutral species.
- (3) In resonance, electrons always flow from higher EN atom to lower EN atom
- (4) CH_3^+ is diamagnetic in nature.

62. Which of the following will not act as $\text{Nu}^\ominus$?

- (1) $\text{CH}_2=\text{CH}_2$ (2) CH_3NH_2
- (3) $:\text{CCl}_2$ (4) H_2O

63. Select the correct order of stability?



- (1) $c > b > a$ (2) $b > a > c$
- (3) $a > b > c$ (4) $a > c > b$

64. Which of the following molecule does not have any inductive effect?

- (1) $\text{H}_3\text{C}-\text{Cl}$ (2) CH_3-NO_2
- (3) $\text{H}_3\text{C}-\text{CH}=\text{CH}_2$ (4) 

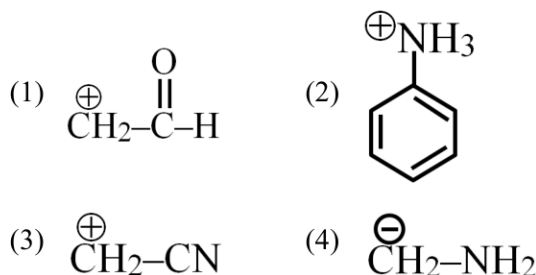
65. Which of the following has maximum pK_b ?

- (1) $\text{CH}_3-\text{CH}_2-\text{NH}_2$ (2) $(\text{CH}_3-\text{CH}_2)_2\text{NH}$
- (3) $(\text{CH}_3\text{CH}_2)_3\text{N}$ (4) All have equal p^{kb}

66. Correct statement?

- (1) Hyperconjugation is dominant over mesomeric effect.
- (2) Hyperconjugation is distance dependent effect.
- (3) Hyperconjugation involves delocalization of πe^-
- (4) Hyperconjugation is a permanent effect.

67. Which of the following molecule has resonance?



68. What will be osmotic pressure of 0.2 M $\text{K}_4[\text{Fe}(\text{CN})_6]$ at 27°C ?

- (1) 24.6 atm (2) 49.2 atm
- (3) 4.92 atm (4) 2 atm

69. A non-ideal solution obtained by mixing 30 mL chloroform and 50 mL acetone will show :-

- (1) $\Delta H = -\text{ve}$ and total volume < 80 mL
- (2) $\Delta H = -\text{ve}$ and total volume $= 80$ mL
- (3) $\Delta H = +\text{ve}$ and total volume > 80 mL
- (4) $\Delta H = +\text{ve}$ and total volume $= 80$ mL

70. The values of observed and theoretical molecular weight of calcium nitrate are respectively 65.6 and 164. The degree of dissociation of calcium nitrate will be :

- (1) 25% (2) 50% (3) 75% (4) 60%

71. Henry's law constants for gases are given below at 293 K. Which one is more soluble in water at that temp. and same pressure:-

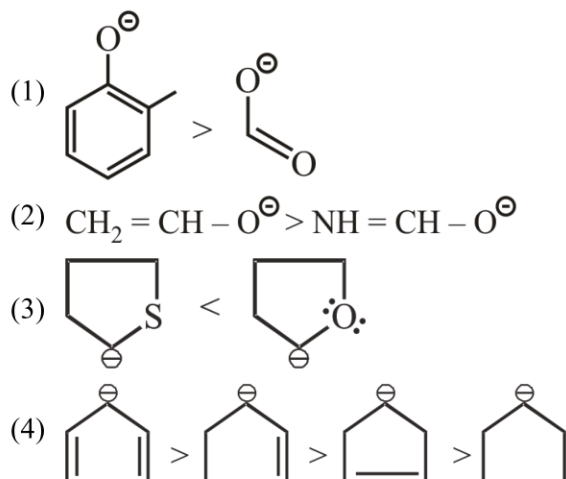
- (1) A ($K_H = 34.86$ Kbar) (2) B ($K_H = 69.16$ Kbar)
- (3) C ($K_H = 144.97$ Kbar) (4) D ($K_H = 88.84$ Kbar)

72. An ideal solution was found to have a vapour pressure of 80 torr when the mole fraction of a non-volatile solute was 0.2. What would be the vapour pressure of the pure solvent at the same temperature ?
- (1) 64 torr (2) 80 torr
(3) 100 torr (4) 400 torr
73. Electrolytic conduction differs from metallic conduction from the fact that in the former
- (1) The resistance increases with increasing temperature
(2) The resistance decreases with increasing temperature
(3) The resistance remains constant with increasing temperature
(4) The resistance is independent of the length of the conductor
74. Correct expression for G^* (cell constant) is :
- (1) $\frac{A}{\ell}$ (2) $\frac{R}{k}$
(3) $\frac{k}{R}$ (4) Rk
75. Relative lowering of vapour pressure of a dilute solution is 0.2. What is the mole fraction of the non-volatile solute ?
- (1) 0.8 (2) 0.5
(3) 0.3 (4) 0.2
76. 30 ml of 2M HCl solution is diluted by 70 ml of H_2O . Molarity of final solution is—
- (1) $\frac{6}{7}M$ (2) 0.6 M
(3) $\frac{7}{6}M$ (4) 1.6 M
77. 6.0 g of urea (molecular mass = 60) was dissolved in 9.9 moles of water. If the vapour pressure of pure water is P^0 , the vapour pressure of solution is :
- (1) $0.10 P^0$ (2) $1.10 P^0$
(3) $0.90 P^0$ (4) $0.99 P^0$
78. The degree of association is 70% for the following reaction $2A \rightarrow (A)_2$. Calculate the Van't Hoff factor.
- (1) 0.65 (2) 1.3
(3) 0.35 (4) 0.85
79. Which of the following statements regarding solubility of gas in water is correct ?
- (1) As temperature increases, solubility of gas also increases.
(2) A more polar gas will have lesser solubility as compared to non-polar gas.
(3) Gases will be more soluble when dissolved at higher pressures.
(4) The dissolution process is always endothermic.
80. For a binary ideal liquid solution, the total pressure of the solution is given as :-
- (1) $P_{\text{total}} = P_A^0 + (P_A^0 - P_B^0)X_B$
(2) $P_{\text{total}} = P_B^0 + (P_A^0 - P_B^0)X_A$
(3) $P_{\text{total}} = P_B^0 + (P_B^0 - P_A^0)X_A$
(4) $P_{\text{total}} = P_B^0 + (P_B^0 - P_A^0)X_B$
81. Mole fraction of a solute in an aqueous solution is 0.2. The molality of the solution will be
- (1) 13.88 (2) 1.388
(3) 0.138 (4) 0.0138
82. If α is the degree of dissociation of $K_4[Fe(CN)_6]$, then abnormal mass of complex in the solution will be :-
- (1) $M_{\text{normal}} (1+2\alpha)^{-1}$ (2) $M_{\text{normal}} (1+3\alpha)^{-1}$
(3) $M_{\text{normal}} (1+\alpha)^{-1}$ (4) $M_{\text{normal}} (1+4\alpha)^{-1}$
83. Maximum elevation in boiling point will be observed in which of the following solution :-
- (1) 0.1 M $CaCl_2$
(2) 0.1 M NaCl
(3) 0.1 M CH_3COONa
(4) 0.1 M H_2NCONH_2

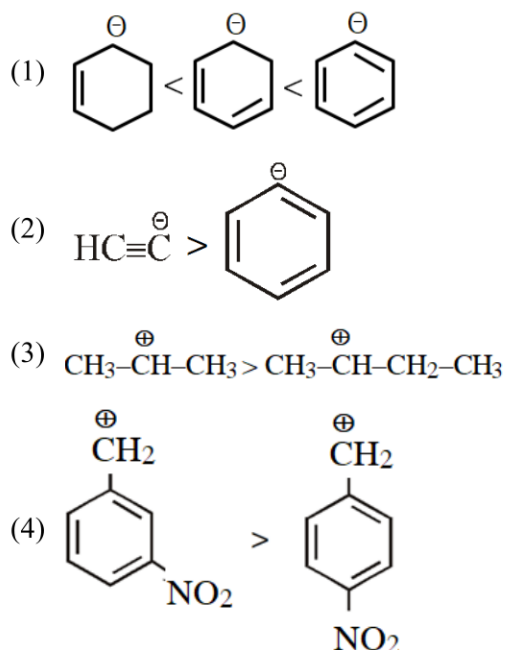
84. Molal elevation constant, in fact is the ratio of elevation in boiling point and :-
- Molarity
 - Molality
 - Mole fraction of solute
 - Mole fraction of solvent
85. The van't Hoff factor i for a dilute solution of $K_3[Fe(CN)_6]$ is :-
- 10
 - 4
 - 5
 - 0.25

SECTION-B (CHEMISTRY)

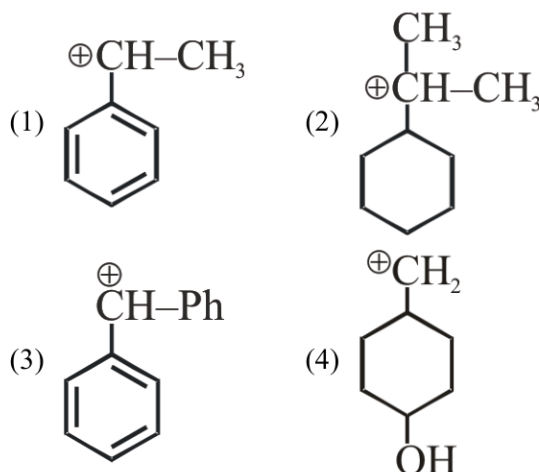
86. Which is correct stability order of following carbanion :-



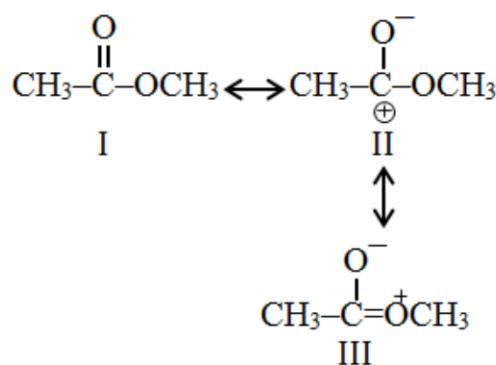
87. Select the incorrect order of stability :-



88. Which of the following is most stable :-

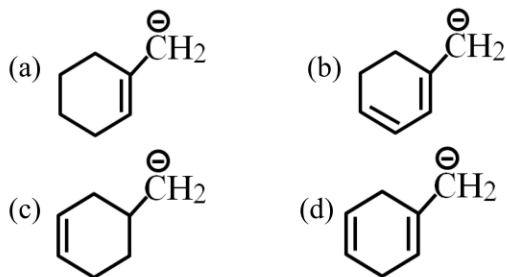


89. Which of the following resonating structure will contribute minimum to resonance hybrid?



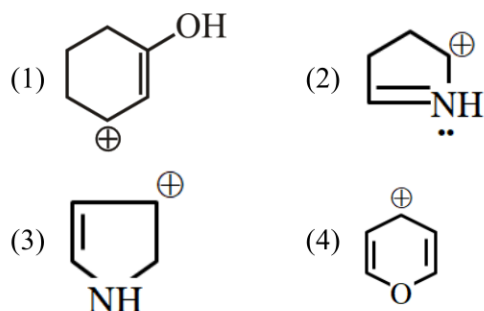
- (1) I
- (2) II
- (3) III
- (4) All structures contribute equally
90. Select the correct statement :-
- The energy of the actual structure of the molecule is higher than that of any of the canonical structures.
 - Resonance energy is the energy difference between the actual structure and the highest energy resonance structure.
 - The more the number of important canonical structures, the more will be the resonance energy.
 - Canonical structures contribute in order of their energy to form the resonance hybrid.

91. Stability order ?

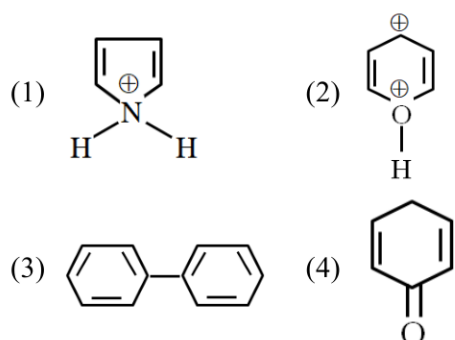


- (1) $a > b > c > d$ (2) $a > d > b > c$
 (3) $b > a > c > d$ (4) $b > d > a > c$

92. Which cation has no resonance with lone pair ?



93. Which of the following is aromatic ?



94. Volume of 0.1 M aqueous KOH required to neutralize 40 ml of 0.1 M aqueous solution of phosphorus acid (H_3PO_3)

- (1) 40 ml (2) 80 ml
 (3) 120 ml (4) 20 ml

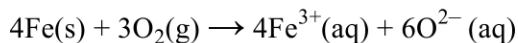
95. Benzene and toluene form nearly ideal solution. At 20°C , the vapour pressure of benzene is 75 torr and that of toluene is 22 torr. The partial vapour pressure of benzene at 20°C for a solution containing 78 g of benzene and 46 g of toluene in torr is :

- (1) 25 (2) 50 (3) 53.5 (4) 37.5

96. Elevation in boiling point was 0.52°C when 6 g of a compound x was dissolved in 100 g of water. Molecular weight of x is : ($K_b = 0.52 \text{ k kg mol}^{-1}$)

- (1) 120 (2) 60
 (3) 100 (4) 342

97. In the reaction :



which of the following statement is incorrect?

- (1) This is redox reaction
 (2) Fe is reducing agent
 (3) O_2 is an oxidizing agent
 (4) Fe is reduced to Fe^{3+}

98. The resistance of 0.01N solution of an electrolyte AB at 328K is 100 ohm. The specific conductance of solution is : (cell constant = 1 cm^{-1})

- (1) $100 \text{ ohm}^{-1} \text{ cm}^{-1}$
 (2) 10^{-2} ohm^{-1}
 (3) $10^{-2} \text{ ohm}^{-1} \text{ cm}^{-1}$
 (4) 10^2 ohm-cm

99. 2 moles each of liquids A and B are dissolved to form an ideal solution. What will be the mole fraction of B in the vapour phase :-
 $p_A^0 = 120 \text{ torr}$; $p_B^0 = 80 \text{ torr}$.

- (1) 1/4 (2) 1/2
 (3) 3/5 (4) 2/5

100. At 323 K, the vapour pressure in millimeters of mercury of a methanol-ethanol solution is represented by the equation $P = 120 X_A + 140$, where X_A is the mole fraction of methanol. Then the value of $\lim_{X_A \rightarrow 1} \frac{P_A}{X_A}$ is :-

- (1) 250 mm
 (2) 140 mm
 (3) 260 mm
 (4) 20 mm

Topic : EVOLUTION (ADAPTIVE RADIATION COMPLETE), GENETICS (TILL TAUGHT), SEXUAL REPRODUCTION IN FLOWERING PLANTS

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SECTION-A (BIOLOGY-I)

101. Which of the following is a vestigial organ ?

- (1) Wisdom tooth (2) Nails
(3) Scalp hairs (4) All of these

102. Archaeopteryx is a connecting link between :

- (1) Reptiles and birds
(2) Fishes and amphibians
(3) Reptiles and amphibians
(4) Mammals and birds

103. Presence of gill slits in the embryos of all vertebrates supports the theory of :-

- (1) Organic evolution
(2) Biogenesis
(3) Metamorphosis
(4) Recapitulation

104. **Assertion :-** Australian marsupials can be taken as an example of adaptive radiation.

Reason :- A number of marsupials, evolved from an ancestral stock, but all within the Australian continent in different habitats.

- (1) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion.
(2) Both Assertion and Reason are correct, but Reason is the NOT correct explanation of Assertion.
(3) Assertion is correct, but Reason is incorrect.
(4) Assertion is incorrect, but Reason is correct.

105. How many features are applicable for Darwinism : Natural selection, Origin of species, Struggle for existence, Use and disuse of organs, Branching descent, Internal vital force, Inheritance of acquired characters

- (1) 1 (2) 3 (3) 4 (4) 5

106. Select the incorrect statements :-

- (A) The essence of Darwinian theory of evolution is natural selection
(B) Evolution is a directed process in the sense of determinism
(C) The work of Darwin on population influenced Malthus.
(D) During evolution the rate of appearance of new forms is linked to the life cycle

- (1) A and B (2) B and C
(3) A and D (4) B, C and D

107. According to Darwinism, long necked giraffes evolved because :

- (1) nature selected only long necked ones
(2) humans preferred only long necked ones
(3) short necks suddenly changed into long necks
(4) of stretching of necks over many generations by short necked ones

108. Which of the following evidences does not favour the lamarckian concept of inheritance of acquired characters ?

- (1) Melanization in peppered moth
(2) Absence of limbs in snakes
(3) Presence of webbed toes in aquatic birds
(4) Lack of pigment in cave-dwelling animals

109. The by-chance change in allelic frequency of a small, isolated population is an example of:

- (1) Gene migration (2) Gene flow
(3) Gene recombination (4) Genetic drift

110. Match the columns I and II, and choose the correct combination from the options given.

	Column I		Column II
(a)	Invertebrates evolved	(p)	65 mya
(b)	Sea weeds and few plants evolved	(q)	200 mya
(c)	Jawless fishes evolved	(r)	320 mya
(d)	Fish like reptiles evolved	(s)	350 mya
(e)	Dinosaurs disappeared	(t)	500 mya

- (1) a-r, b-t, c-s, d-p, e-q (2) a-s, b-r, c-t, d-q, e-p
(3) a-t, b-s, c-r, d-p, e-q (4) a-t, b-r, c-s, d-q, e-p

111. **Assertion** : When reptiles came down, mammals took over the earth.

Reason : Mammals were viviparous and more intelligent.

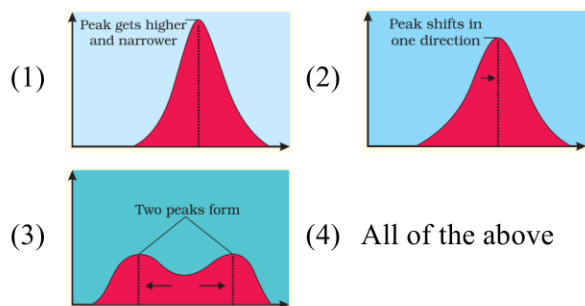
- (1) Both Assertion & Reason are True & the reason is correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are false.

112. **Statement-I** : About 65 million years ago suddenly dinosaurs reappeared on the earth.

Statement-II : Tyrannosaurus rex was about 200 feet in height and had huge fearsome daggerlike teeth.

- (1) Statement I and II both are correct
- (2) Statement I and II both are incorrect
- (3) Only Statement I is correct
- (4) Only Statement II is correct

113. Which graph represent stabilising selection.



114. Which is the most important factor for evolution of new species :-

- (1) Reproductive isolation
- (2) Geographical isolation
- (3) Adaptation
- (4) Natural selection

115. Change in allelic frequency at species or below species level can be termed as :

- (1) Mega evolution
- (2) Macro evolution
- (3) Micro evolution
- (4) Silent evolution

116. Cave paintings by Pre-historic humans can be seen at _____ rock shelter in _____ district of _____.

- (1) Raisen, Bhimbetka, Madhya Pradesh
- (2) Raisen, Bhimbetka, Rajasthan
- (3) Bhimbetka, Raisen, Madhya Pradesh
- (4) Bhimbetka, Raisen, Uttar Pradesh

117. Match the organism with its correct cranial capacity.

Column-I		Column-II	
A.	Neanderthal man	(i)	1600 cc
B.	Modern man	(ii)	900 cc
C.	Cromagnon man	(iii)	1400 cc
D.	Java man	(iv)	1350 cc

- (1) A-(iv), B-(iii), C-(i), D-(ii)
- (2) A-(iv), B-(i), C-(iv), D-(ii)
- (3) A-(iii), B-(i), C-(ii), D-(iv)
- (4) A-(iii), B-(iv), C-(i), D-(ii)

118. Fossils of man-like bones discovered in Ethiopia and Tanzania belonged to -

- A. Homo habilis
 - B. Homo erectus
 - C. Australopithecus
 - D. Ramapithecus
- (1) C only
 - (2) A and C
 - (3) B and C
 - (4) A, C and D

119. How many of the following statements are true?

- A. About 3-4 mya, man-like primates walked in eastern Africa.
- B. Australopithecus were probably taller than 4 feet but walked up right.
- C. Two mya, Australopithecines probably lived in East African grasslands.
- D. Australopithecines used hides to protect their body and buried their dead.

- (1) Four
- (2) Three
- (3) Two
- (4) One

120. Which of the following was about 20 feet in height?
- (1) Brachiosaurs (2) Triceratops
(3) Tyrannosaurus rex (4) Ichthyosaurs
121. Which of the following statement is not correct?
- (1) Evolution of horse, elephant, dog, etc. are special stories of evolution.
(2) The most successful story is the evolution of man with language skills and self-consciousness.
(3) Pre-existing advantageous mutations when selected will result in observation of new phenotypes
(4) Natural selection is based on certain observations which are fictional.
122. First organisms to invade land were :-
- (1) Coelacanth (2) Amphibians
(3) Frogs (4) Plants
123. Evolution of different species from one is called :-
- (1) Divergent evolution
(2) Convergent evolution
(3) Adaptive radiation
(4) Adaptive convergence
124. Original idea of "survival of fittest" was proposed by-
- (1) Charles Darwin (2) Herbert Spencer
(3) Hugo deVries (4) Hardy-Weinberg
125. In a particular geographical area, population of a species was drastically decreased due to natural calamities and few genes/alleles were permanently lost but survivors reproduced and gradually restored the population.
Above case represents-
- (1) Founder effect (2) Gene migration
(3) Gene flow (4) Bottleneck effect
126. Which plate was prepared by the help of velvet covered wooden block in Lederberg's replica plate experiment?
- (1) Master plate (2) Replica plate
(3) Wooden plate (4) Velvet plate
127. In a particular animal species, individual with long forelimbs survived by eating tree leaves and individuals with short forelimbs survived by eating grass. However, individuals with medium length forelimbs survived less as they faced difficulties in eating both grass and tree leaves.
Which of the following is represented by above case?
- (1) Balancing selection
(2) Directional selection
(3) Disruptive selection
(4) Stabilizing selection
128. Ultimate source of variation is?
- (1) Mutations
(2) Gene recombination
(3) Hybridization
(4) Gene migration
129. Evolution of horse represents which type of speciation?
- (1) Allopatric (2) Sympatric
(3) Quantum (4) Phyletic
130. Which of the following is not an example of natural selection?
- (1) Drug resistance in microbes
(2) Industrial Melanism
(3) Sickle cell anemia and malaria
(4) Domestic Pigeons
131. In England, which variant of *Biston betularia* was completely wiped out in 1920s?
- (1) White winged (2) Dark winged
(3) Neither 1 nor 2 (4) Both 1 and 2

132. Tiger cat shows convergence with :-

- (1) Tiger
- (2) Cat
- (3) Bobcat
- (4) Catfish

133. According to Darwin; fitness is due to characters which are :-

- (1) Genetic
- (2) Inbuilt
- (3) Acquired
- (4) Lost

134. Single step large mutation is called :-

- (1) Saltatory conduction
- (2) Saltation
- (3) Long jump
- (4) Gene recombination

135. Which of the following statement is incorrect?

- (1) Habitat fragmentation and genetic drift may accentuate variations leading to appearance of new species and hence evolution.
- (2) Homology is accounted for by the idea of branching descent.
- (3) There has been parallel evolution of human brain and language.
- (4) Interspecific struggle is always most potent force for organic evolution.

SECTION-B (BIOLOGY-I)

136. Atavism in man means

- (1) Appearance of new characters
- (2) Evolution of existing characters
- (3) Reappearance of ancestral characters
- (4) Loss of some pre-existing characters

137. **Assertion** : Mammals have evolved from egg laying species.

Reason : Prototherians are oviparous.

- (1) Both Assertion & Reason are True & the reason is correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are false.

138. Select the set of animals showing convergent evolution with respect to each other.

- (1) Koala and Bandicoot
- (2) Koala and Kangaroo
- (3) Lemur and Spotted cuscus
- (4) Bobcat and Flying squirrel

139. What is the correct arrangement of periods of palaeozoic era in ascending order in geological time scale?

- (1) Cambrian → Devonian → Ordovician → Silurian → Carboniferous → Permian
- (2) Cambrian → Ordovician → Silurian → Devonian → Carboniferous → Permian
- (3) Cambrian → Ordovician → Devonian → Silurian → Carboniferous → Permian
- (4) Silurian → Devonian → Cambrian → Ordovician → Permian → Carboniferous

140. Consider the following four statements (a-d) and select the option which includes all the **correct** ones only.

- (a) Evolution is a process
 - (b) Evolution is consequence of a process
 - (c) Natural resources are limited
 - (d) Populations are stable in size except for seasonal fluctuation
- (1) Statements a, b, c (2) Statements a, b, c and d
(3) Statements a, b, d (4) Statements a and d

- 141. Statement-I :** Hugo-de vries believed that it is mutation which causes evolution and not the minor variations.
- Statement-II :** Mutations are random and directionless.
- (1) Statement I and II both are correct
 - (2) Statement I and II both are incorrect
 - (3) Only statement I is correct
 - (4) Only statement II is correct
- 142.** According to Neo-darwinism theory which one of the following is not a source of genetic variation?
- (1) Genetic drift
 - (2) Hybridization
 - (3) Mutation
 - (4) Competition
- 143.** Populations are said to be sympatric when
- (1) two populations live together and freely interbreed to produce sterile offspring.
 - (2) two populations are physically isolated by natural barriers.
 - (3) two populations are isolated but occasionally come together to interbreed.
 - (4) two populations share the same environment but cannot interbreed.
- 144.** Two population of "deer mouse" are present. One in the middle of the forest and the other from a nearest crop field habitat. The two populations show no interbreeding naturally but can interbreed in laboratory. This type of reproductive isolation is called :-
- (1) Seasonal isolation
 - (2) Ecological isolation
 - (3) Physiological isolation
 - (4) Behavioural isolation
- 145.** Read the following statements w.r.t. human evolution.
- (a) Pre-historic cave art developed about 18,000 years ago
 - (b) Agriculture was started by modern man.
 - (c) Domestication of animals was started by cromagnon man.
 - (d) Ramapithecus was more man like while Dryopithecus was more ape like.
- How many statements are not correct ?
- (1) Zero (2) One (3) Two (4) Three
- 146.** Skull shown below is of :-
- 
- (1) Adult human (2) Baby chimpanzee
 - (3) Baby human (4) Adult chimpanzee
- 147.** Which of the following resulted in beginning of human settlements?
- (1) Agriculture (2) Cave art
 - (3) Hunting (4) Use of fire
- 148.** Zosterophyllum and Rhynia-type plants evolved from :-
- (1) Tracheophyte ancestors
 - (2) Psilophytons
 - (3) Bryophytes
 - (4) Herbaceous lycopods
- 149.** Darwin's finches represents how many of the following phenomenon?
Adaptive radiation, divergent evolution, sympatric speciation, natural selection, anthropogenic action, homology, founder's effect.
- (1) Three (2) Four (3) Five (4) Six
- 150.** Which of the following primate is closest to humans?
- (1) Chimpanzee (2) Lemur
 - (3) Neanderthal man (4) Handy man

Topic : EVOLUTION (ADAPTIVE RADIATION COMPLETE), GENETICS (TILL TAUGHT), SEXUAL REPRODUCTION IN FLOWERING PLANTS

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SECTION-A (BIOLOGY-II)

151. How many gametes are present in fully developed female gametophytes of *Capsella* ?

- (1) 8 (2) 7
(3) 2 (4) 1

152. What is common about *Capsella* and *Pinus* ?

- (1) Gametes are non motile
(2) Both exhibit double fertilisation
(3) They are vascular plants
(4) Both (1) and (3)

153. What is true about tapetum ?

- (1) It is known as nutritive layer.
(2) It is binucleated or multinucleated.
(3) It is inner most layer of microsporangial wall
(4) All of them

154. Which one is entomophilous plant ?

- (1) *Zostera* (2) *Hydrilla*
(3) Water hyacinth (4) *Vallisneria*

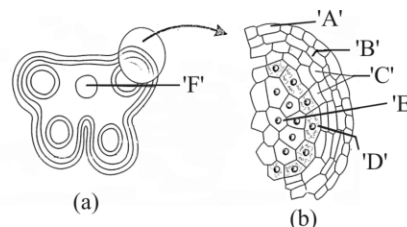
155. The wall of anther consist four different layers where :-

- (1) Middle layer lies between endothecium and Tapetum
(2) Endothecium lies inner to middle layer
(3) Tapetum lies just below the endothecium
(4) Tapetum lies just below the epidermis

156. Hilum is ?

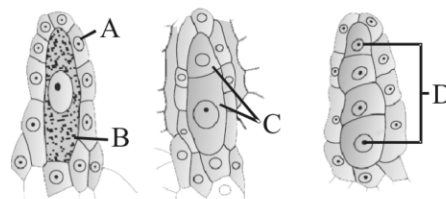
- (1) Region between placenta and ovule
(2) Region between funicle and body of ovule
(3) Region between embryosac and integument
(4) Found between two microsporangia of a anther

157. Given figure 'a' is transverse section of anther and diagram 'b' is enlarged view of one microsporangium showing wall layers. In which labelled structures A to F are :



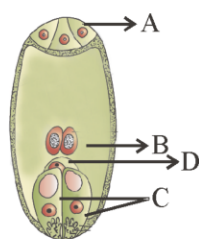
- (1) A → Epidermis, B → Exothecium, C → Middle layer, D → Tapetum, E → Microspore mother cells, F → Connective
(2) A → Epidermis, B → Endothecium, C → Middle layer, D → Tapetum, E → Microspore mother cells, F → Connective
(3) A → Endothecium, B → Epidermis, C → Tapetum, D → Middle layer, E → Connective, F → Microspore mother cells
(4) A → Middle layer, B → Epidermis, C → Endothecium, D → Tapetum, E → Microspore mother cells, F → Connective

158. Given below is the diagrammatic view of different stages during embryo sac development. Identify the parts labelled A, B, C and D and select the right option about them.

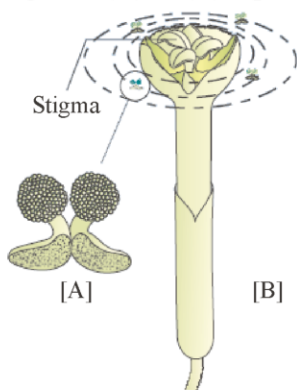


	Part-A	Part-B	Part-C	Part-D
(1)	Nucellus	Megaspore mother cell	Megaspore tetrad	Megaspore dyad
(2)	Nucellus	Megaspore mother cell	Megaspore dyad	Megaspore tetrad
(3)	Megaspore mother cell	Nucellus	Megaspore dyad	Megaspore tetrad
(4)	Megaspore mother cell	Nucellus	Megaspore tetrad	Megaspore dyad

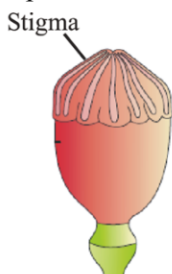
159. Select the incorrect matching with respect to following diagram :-



- (1) A-haploid structure, degenerate after syngamy
 - (2) B-diploid structure, forms embryo sac after fertilization
 - (3) C-one degenerate before syngamy
 - (4) D-undergoes syngamy during double fertilization
160. Following diagrams (A) & (B) represent ?



- (1) (A) Female flower of *Hydrilla*
(B) Male flower of *Hydrilla*
 - (2) (A) Male flower of *Hydrilla*
(B) Female flower of *Hydrilla*
 - (3) (A) Female flower of *Vallisneria*
(B) Male flower of *Vallisneria*
 - (4) (A) Male flower of *Vallisneria*
(B) Female flower of *Vallisneria*
161. Following figure represents :-



- (1) Multicarpellary, syncarpous gynoecium of *Papaver*
 - (2) Multicarpellary, syncarpous gynoecium of *Michelia*
 - (3) Multicarpellary, apocarpous gynoecium of *Papaver*
 - (4) Multicarpellary, apocarpous gynoecium of *Michelia*
162. Given below are two statements :
- Assertion (A) :-** In over 60 percent of angiosperms, pollen grains are shed at 3-celled stage.
- Reason (R) :-** The mature pollen grain contains two cells, the vegetative cell and generative cell.
- (1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
 - (2) (A) is correct but (R) is not correct
 - (3) (A) is not correct but (R) is correct
 - (4) Both (A) and (R) are correct and (R) is the correct explanation of (A)
163. Given below are two statements :-
- Statement-I :** Geitonogamy is functionally cross-pollination involving a pollinating agent.
- Statement-II :** Geitonogamy is genetically similar to autogamy since pollen grains come from the same plant.
- Choose correct answer from the options given below :-
- (1) Both statement-I and statement-II are incorrect.
 - (2) Statement-I is correct but statement-II is incorrect.
 - (3) Both statement-I and statement-II are correct.
 - (4) Statement-I is incorrect but statement-II is correct.
164. Monoecious plant, prevents :-
- (1) Autogamy
 - (2) Geitonogamy
 - (3) Both autogamy and geitonogamy
 - (4) Pollination is not possible
165. In *Vallisneria* :-
- (1) Both male and female flowers break from the plant and float on the surface of water.
 - (2) Only the female flowers break from the plant, while the male flowers are brought to the surface by long stalks.
 - (3) Only the male flowers break from the plant and rise to the surface, while the female flowers are brought to the surface by long pedicels (stalks).
 - (4) Any of the two types of flowers can break.

166. Match the column :-

	Column-I	Column-II
i	Pollenkitt	(a) <i>Cuscuta</i>
ii	Pollinium	(b) <i>Calotropis</i>
iii	Parasite plant	(c) <i>Capsella</i>
iv	Insect and wind pollination	(d) Water hyacinth

- (1) i-c, ii-b, iii-a, iv-d (2) i-c, ii-a, iii-d, iv-b
(3) i-d, ii-b, iii-a, iv-c (4) i-c, ii-d, iii-a, iv-b

167. Wind pollinated plants differ from insect pollinated plants in having :-

- (1) small petals and sticky pollen
(2) coloured petals and heavy pollen
(3) coloured petals and nectar
(4) without nectar and light weight pollens

168. Read the following five statements (A–E) and answer as asked next to them :-

- (A) Xenogamy is the only type of pollination, which brings genetically different types of pollen grains to the stigma
(B) The basal part of ovule is known as nucellus
(C) The number of ovules are many in papaya and watermelon
(D) Stored pollen grains can be used in crop breeding programmes
(E) Generative cell is small and float in the cytoplasm of vegetative cell
How many of the above statements are wrong ?

- (1) Four (2) Three (3) Two (4) One

169. Find out the correct match from the following table :-

Column-I	Column-II	Column-III
(i) Nucellus	Perisperm	Diploid
(ii) Micropyle	Ovule	Entry of water
(iii) Synergid	Filiform apparatus	Chemotropic movement of pollen tube

- (1) (i) only (2) (i) and (ii)
(3) (iii) and (ii) (4) (i) and (iii)

170. The number of mitotic division required to form a mature pollen grain in the flowering plants is-

- (1) One (2) Two
(3) Three (4) Four

171. The types of flowers which always produce seeds even in the absence of pollinators-

- (1) Chasmogamous flowers
(2) Bisexual flowers
(3) Cleistogamous flowers
(4) Unisexual flowers

172. Choose the wrong one regarding gynoecium.

- (1) Gynoecium represents the female reproductive part of flower
(2) The gynoecium may be syncarpous or apocarpous
(3) The number of ovules in orchids is one
(4) The ovules are attached to placenta

173. Feathery stigma and well - exposed stamens are characteristic of-

- (1) Wind pollinated flowers
(2) Water pollinated flowers
(3) Insect pollinated flowers
(4) Bat pollinated flowers

174. Mark the incorrect statement-

- (1) Outer layer of anther wall are protective in function
(2) Sporogenous tissue, occupies the centre of each microsporangium
(3) Cells of tapetum show meiosis
(4) Ploidy level of microspore tetrad is haploid

175. Which of the following has proved helpful in preserving pollen as fossils?

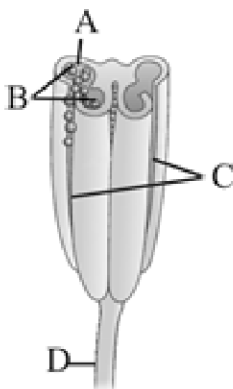
- (1) Pollenkitt (2) Cellulosic intine
(3) Sporopollenin (4) Pellicle

176. In papaya male and female flowers are present on different plants. It permits-
- Autogamy
 - Geitonogamy
 - Both autogamy and geitonogamy
 - Xenogamy
177. An angiospermic plant is having 24 chromosomes in its leaf cells. The number of chromosomes present in synergid, pollen grain, nucellus & endosperm will be respectively ?
- 12,12,12, 72
 - 8, 8,12, 36
 - 12,12, 24, 36
 - 12,12,12, 36
178. A typical angiosperm anther is _____ with each lobe having _____ theca i.e. they are _____
- Bilobed, two, ditheous
 - Ditheous, two, bilobed
 - Bilobed, four, ditheous
 - Ditheous, four, bilobed
179. Which plant is perennial but monocarpic ?
- Bamboo
 - Strobilanthus*
 - Mango
 - Both (1) & (2)
180. In embryo-sac polar nuclei are situated :
- Below the antipodals
 - Above the egg apparatus
 - Below the egg apparatus
 - inside the synergids
181. How many divisions are required to form female gametophyte from MMC ?
- One meiosis ; two mitosis
 - Two meiosis ; three mitosis
 - One meiosis ; three mitosis
 - One meiosis ; One mitosis
182. Pollination by _____ is more common amongst abiotic pollinations.
- water
 - wind
 - Insects
 - Birds
183. Which of the following is not a monoecious plant ?
- Castor
 - Maize
 - Papaya
 - Coconut
184. Choose incorrect statement-
- The number of ovules in an ovary is one in paddy
 - The number of ovules in an ovary is many in papaya
 - Lotus has syncarpous gynoecium
 - Wheat mango consist of one ovule
185. Hypohydrophily found in :
- Zostera*
 - Hydrilla*
 - Vallisneria*
 - Both (1) & (2)

SECTION-B (BIOLOGY-II)

186. Triple fusion in angiosperm is the fusion of second sperm with :
- Antipodal cell and one synergid cell
 - Two antipodal cells
 - Two synergid cells
 - Two polar nuclei
187. Which of the following phenomenon is not related to pollen tube ?
- Apical growth
 - Chemotactic movement
 - Siphonogamy
 - Nonmotile gametes

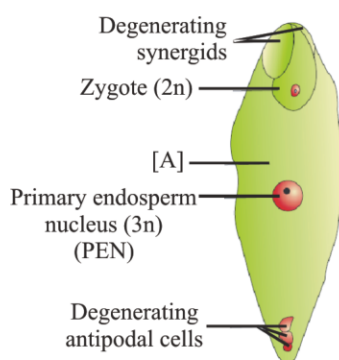
188. Given below is the diagrammatic view of three dimensional cut section of an anther. Identify the parts labelled A, B, C and D and select the right option about them.



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	Part-A	Part-B	Part-C	Part-D
1	Pollen sacs	Pollen grains	Filament (Stalk)	Line of dehiscence
2	Pollen grains	Pollen sacs	Line of dehiscence	Filament (Stalk)
3	Pollen grains	Pollen sacs	Filament (Stalk)	Line of dehiscence
4	Pollen sacs	Line of dehiscence	Pollen grains	Filament (Stalk)

189. Following diagram represents fertilized embryo sac. Identify labelled structure [A] of this diagram:-



- (1) Embryo
- (2) Perisperm
- (3) Primary endosperm cell
- (4) Central cell

190. **Assertion :-** Double fertilization and triple fusion is the specific or universal characteristic of angiosperm.
Reason :- Five nuclei and two gametes participate in double fertilization.

- (1) Both assertion & reason are true & the reason is a **correct** explanation of the assertion.
- (2) Both assertion & reason are true but reason is **not** a correct explanation of the assertion.
- (3) Assertion is **true** but the reason is **false**
- (4) Both assertion & reason are **false**.

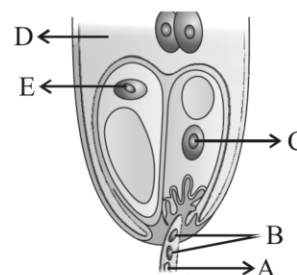
191. Consider the following four statements (a-d) and select the option which includes all the correct ones only.

- (a) All aquatic plants use water for pollination.
- (b) Not all aquatic plants use water for pollination.
- (c) *Vallisneria* flowers are pollinated by water under the water surface.
- (d) Water hyacinth and water lily are pollinated by insects.

Option :-

- (1) Statements (b), (c) and (d)
- (2) Statements (b) and (d)
- (3) Statements (a), (c) and (d)
- (4) Statements (a) and (c)

192. Identify the plants parts labelled A, B, C, D and E in the figure given below and select the right option.



	A	B	C	D	E
1	Male gametes	Vegetative nucleus	Central cell	Egg	Synergid
2	Generative nucleus	Male gamete	Central cell	Synergid	Egg
3	Male gametes	Vegetative nucleus	Synergid	Central cell	Egg
4	Vegetative nucleus	Male gamete	Synergid	Central cell	Egg

- 193.** Pollen grains are generally _____ in outline measuring _____ micrometers in diameter.
- (1) Spherical, 25-50 (2) Oblong, 25-50
(3) Oval, 10-25 (4) Spherical, 75-100
- 194.** The vegetative cell is-
- (1) Small, has large irregularly shaped nucleus
(2) Large, has large irregular shaped nucleus
(3) Large with spindle shaped nucleus
(4) Small, spindle shaped nucleus
- 195.** Which is incorrect statement?
- I. Each cell of sporogenous tissue in anther is capable of giving rise to microspore tetrad.
II. The pollen grain represent male gametophyte.
III. Pollen grains are usually multicellular.
IV. Sporopollenin is one of the most resistance organic material which can be destroyed only by strong acids and alkali.
- (1) I, II are incorrect but III, IV are correct
(2) III, IV are incorrect but I, II are correct
(3) I, III are incorrect but II, IV are correct
(4) II, IV are incorrect but I, III are correct
- 196.** Which statement is incorrect?
- (1) Intine is the inner wall of pollen grain and exhibit fascinating array of patterns and designs
(2) Female gametophyte has 7 - cells and 8 - nuclei
(3) Carrot grass pollens cause pollen allergy
(4) Pollen grains of pea and rose maintain viability for months
- 197.** Arrange microsporangial wall in sequence of outside to inside-
- (1) Epidermis, middle layer, endothecium, tapetum
(2) Epidermis, endothecium, middle layer, tapetum
(3) Epidermis, middle layer, tapetum, endothecium
(4) Endothecium, middle layer, tapetum, epidermis
- 198.** Choose correct about pollen grain wall-
- (i) It has prominent two layered wall
(ii) Hard outer layer wall called exine
(iii) Exine is composed of sporopollenin
(iv) Sporopollenin form continuous exine
- (1) i, ii, iii, iv (2) i, ii, iii
(3) i, iii only (4) i & iv
- 199.** Sporopollenin is absent in -
- (1) Intine (2) Germ pore
(3) Exine (4) Both (1) & (2)
- 200.** When pollen grain mature-
- (1) It consist of two cell that are two male gamete only
(2) It consist of two cell that are generative & vegetative cell
(3) It consist of two cell that are two male gamete arise from vegetative cell and one generative cell
(4) It consist of three cell that are two male gamete develop meiotically from generative cell and one vegetative cell



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